## 0.1 Preliminaries

### 0.1.1 Definitions

**Definition 0.1.1.** Let $X, Y$ be non-empty sets.
A subset $f \subseteq X \times Y$ is called a *function* from $X$ into $Y$ if it satisfies:

For any $x \in X$, there exists a unique $y \in Y$ such that $(x, y) \in f$

If $f$ is a function on $X$ into $Y$, denote that

$$(x, y) \in f \iff y = f(x)$$

Briefly, a function $f$ from $X$ into $Y$ is denoted as a form:

$$f : X \to Y : x \mapsto y_x$$

where $y_x \in Y$ is determined uniquely by $x$.

Alternatively, the uniqueness condition can be expressed as follows:

$$(x, y) \in f \text{ and } (x, z) \in f \implies y = z$$

**Definition 0.1.2.** Let $f : X \to Y$ be a function, and $A \subseteq X$, $B \subseteq Y$ be subsets.

1. The set $X$ is called a *domain* of $f$.
2. The set $Y$ is called a *codomain* of $f$.
3. The set $f[X] \overset{\text{def}}{=} \{f(x) \in Y \mid x \in X\}$ is called a *range* of $f$. (It is often denoted by $\operatorname{Im} f$.)
4. The set $f[A] \overset{\text{def}}{=} \{f(x) \in Y \mid x \in A\}$ is called an *image* of $A$ under $f$.
5. The set $f^{-1}[B] \overset{\text{def}}{=} \{x \in X \mid f(x) \in B\}$ is called a *preimage* of $B$ under $f$.

**Definition 0.1.3.** Let $f : X \to Y$ be a function.
A map $f$ is called *injective* (or 1-1) if: $f(x_1) = f(x_2) \implies x_1 = x_2$.
A map $f$ is called *surjective* (or onto) if: $f[X] = Y$.
If $f$ is injective and surjective, then it is called *bijective*.

### 0.1.2 Properties

**Lemma 0.1.4.** Let $f : X \to Y$ be a function.

1. There exists $g : Y \to X$ such that $\forall x \in X,\ (g \circ f)(x) = x$ *if and only if* $f$ is injective.
2. There exists $h : Y \to X$ such that $\forall y \in Y,\ (f \circ h)(x) = x$ *if and only if* $f$ is surjective.

That is, $f$ has a right inverse if and only if $f$ is injective, and $f$ has a left inverse if and only if $f$ is surjective.

**Lemma 0.1.5.** Let $f : X \to Y$ be a set, $A_i \subseteq X,\ (i \in I)$, $B_j \subseteq Y,\ (j \in J)$.

1. $f\left[\bigcup_{i \in I} A_i\right] = \bigcup_{i \in I} f[A_i]$
2. $f\left[\bigcap_{i \in I} A_i\right] \subseteq \bigcap_{i \in I} f[A_i]$
3. $f^{-1}\left[\bigcup_{j \in J} B_j\right] = \bigcup_{j \in J} f^{-1}[B_j]$
4. $f^{-1}\left[\bigcap_{j \in J} B_j\right] = \bigcap_{j \in J} f^{-1}[B_j]$

**Lemma 0.1.6.** Let $X, Y, Z$ be sets, $f : X \to Y$ and $g : Y \to Z$ be functions.
For subsets $A \subseteq X$ and $C \subseteq Z$, the following hold:

1. $g[f[A]] = (g \circ f)[A]$
2. $f^{-1}[g^{-1}[C]] = (g \circ f)^{-1}[C]$

**Lemma 0.1.7.** Let $f : X \to Y$ be a function. For subsets $A, B \subseteq X$ and $C, D \subseteq Y$, the following hold:

1. If $A \subseteq B$, then $f[A] \subseteq f[B]$.
2. If $C \subseteq D$, then $f^{-1}[C] \subseteq f^{-1}[D]$.

**Lemma 0.1.8.** Let $f : X \to Y$ be a function. For subsets $A \subseteq X$ and $B \subseteq Y$, the following hold:

1. $f^{-1}[f[A]] \supseteq A$, and equality holds if $f$ is injective.
2. $f[f^{-1}[B]] \subseteq B$, and equality holds if $f$ is surjective.
3. $X \setminus f^{-1}[B] = f^{-1}[Y \setminus B]$
4. $f[X] \setminus f[A] \subseteq f[X \setminus A]$, and equality holds if $f$ is injective.